

# Nikhil Rout

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in nikhil-rout    🌐 NikhilRout    📄 NikhilRout.github.io

## Education

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| <b>Vellore Institute of Technology, Chennai Campus</b>                   | 2022 – 2026   |
| <i>Bachelor of Technology, Electronics and Communication Engineering</i> | CGPA: 9.11/10 |

## Research Experience

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| <b>Research Intern</b> 🌐        | Remote             |
| Vortex GPGPU, ORCAS Lab, UCLA 📄 | May 2025 – Present |

- Developed a Tensor Core Unit (TCU) extension for the open-source RISC-V based Vortex GPGPU, enabling efficient WMMA operations for deep learning workloads
- Architected and implemented a high-performance configurable mixed-precision Fused Dot Product (FEDP) unit as part of the TCU in SystemVerilog
- Extended custom float conversions, SGEMM test, and cycle-level simulator implementation for FP8/BF8 support in C++
- Introduced a novel fused pipeline strategy for supporting Integer arithmetic within the floating-point datapath, maximizing component reuse while incurring minimal overhead
- Performed microarchitecture design space explorations, evaluating 3:2 vs 4:2 compressors in Carry-Save Adders, and format-wise Dedicated vs Shared multiplier schemes
- Reduced dynamic power consumption through clock/enable gating across FEDP pipeline stages during operation in 2:4 structured sparsity mode
- Contributed to TCU specific CI/CD and Verilator-based unittesting verification framework
- Achieved 362.2 MHz operation with 4-cycle latency on the Xilinx U55C FPGA, delivering 5.795 GFLOPS ideal throughput—4.3× the open-source baseline Berkeley HardFloat

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| <b>Summer Research Intern</b> 🌐                           | Remote               |
| Centre for Nanoelectronics and VLSI Design, VIT Chennai 📄 | May 2024 – July 2024 |

- Implemented pipelined radix-2 FFT modules using both DIT and DIF approaches
- Developed Number Theoretic Transform (NTT) modules for efficient polynomial multiplication in lattice-based post-quantum cryptography schemes on FPGAs

## Research Publications

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**N. Rout**, B. Tine, "A Configurable Mixed-Precision Fused Dot Product Unit for GPGPU Tensor Computation," *Vortex Workshop and Tutorials, MICRO 2025*. [Preprint](#) 📄 🌐

**N. Rout**, J.J.J. Nesam, "Optimizing RGB to Grayscale, Gaussian Blur and Sobel-Filter operations on FPGAs for reduced dynamic power consumption," *Best Paper Award, AIIoT 2024*. [IEEE Xplore](#) 📄 🌐

## Projects

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### Carry-Save FAN modification (SIGMA DNN Accelerator)

- Optimized Forward-Adder-Network (FAN) critical path by extending a Carry-Save-Adder tree structure for multi-vector reduction in sparse and irregular GEMM workloads
- Performed Systolic Array Processing Element integration and synthesis PPA analysis

### Flash-Attention CUDA MegaKernel Inference Optimization

- Implemented a CUDA Megakernel for Flash-Attention inference with baseline and Tensor Core variants, leveraging shared memory tiling and online softmax to reduce HBM bottleneck
- Profiled kernel performance on NVIDIA T4 GPU, demonstrating speedup from operator fusion and memory hierarchy optimization

## Skills and Competencies

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**Languages:** Verilog/SystemVerilog, Chisel HDL, C/C++, CUDA, Python, TCL

**Tools:** Quartus Prime, Xilinx Vivado/Vitis, Synopsys VCS/DC, CMake, Git, LaTeX

**Relevant Coursework:** Digital Systems Design, VLSI System Design, Computer Architecture, FPGA-based System Design, Heterogeneous Computing Systems, Deep Learning

## References

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**Dr. Blaise-Pascal Tine**  
*ORCAS Lab, Assistant Professor*  
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**UCLA**

**Dr. Jean Jenifer Nesam J**  
*Assistant Professor*  
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**VIT Chennai**

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**VIT Chennai**